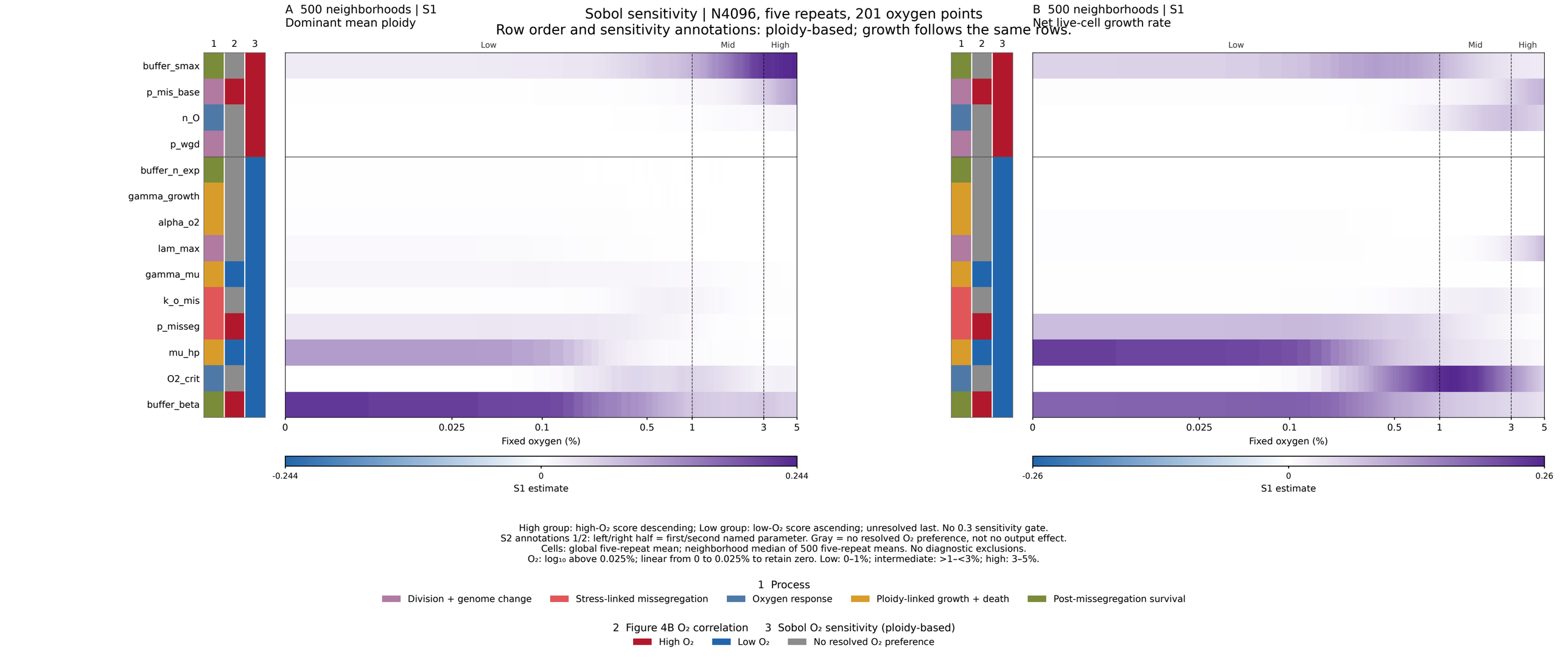
Sobol sensitivity | N4096, five repeats, 201 oxygen points
Row order and sensitivity annotations: ploidy-based; growth follows the same rows.
A 500 neighborhoods | S1
Dominant mean ploidy
B 500 neighborhoods | S1
Net live-cell growth rate
1
2
3
Low
Mid
High
buffer_smax
p_mis_base
n_O
p_wgd
buffer_n_exp
gamma_growth
alpha_o2
lam_max
gamma_mu
k_o_mis
p_misseg
mu_hp
O2_crit
buffer_beta
0
0.025
0.1
0.5
1
3
5
Fixed oxygen (%)
-0.244
0
0.244
S1 estimate
-0.26
0.26
High group: high-O₂ score descending; Low group: low-O₂ score ascending; unresolved last. No 0.3 sensitivity gate.
S2 annotations 1/2: left/right half = first/second named parameter. Gray = no resolved O₂ preference, not no output effect.
Cells: global five-repeat mean; neighborhood median of 500 five-repeat means. No diagnostic exclusions.
O₂: log₁₀ above 0.025%; linear from 0 to 0.025% to retain zero. Low: 0–1%; intermediate: >1–<3%; high: 3–5%.
1 Process
Division + genome change
Stress-linked missegregation
Oxygen response
Ploidy-linked growth + death
Post-missegregation survival
2 Figure 4B O₂ correlation
3 Sobol O₂ sensitivity (ploidy-based)
High O₂
Low O₂
No resolved O₂ preference